

Elijah Shaw-Rutschman

Staff-Level Software Engineer · Platform, Infrastructure & Developer Experience · Open-Source Systems (Python & Nim)

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SUMMARY

Platform and infrastructure engineer with nearly two decades of experience, including 12 years at StyleSeat. Most recently focused on platform, payments, reliability, and developer tooling, including the company's move to AI-assisted engineering. Also an active open-source author in Nim and Python, with well-regarded libraries in lock-free concurrency, compiler tooling, and realtime audio. Earned StyleSeat's highest performance rating ("Exceptional") and was on the formal Staff Engineer promotion track.

SKILLS

Languages

Expert
Python · Nim · TypeScript · Cython · C · SQL

Backend

Expert
Django · Celery · REST APIs

Data & Infrastructure

Advanced
MySQL · Redis · AWS · Terraform · Docker · CircleCI · GitHub Actions

Systems & Low-Level

Advanced
Lock-free concurrency · Memory reclamation · Realtime audio / DSP · Low-level protocols · C / Cython interop

Compiler & Language Tooling

Advanced
libclang · C/C++ header parsing · Cython codegen · Cross-compilation · QEMU multi-arch CI

Payments

Expert
Stripe billing · Stripe subscriptions · Stripe Radar · Webhooks · Fraud detection

AI & LLM Tooling

Advanced
AI-assisted engineering · LLM application development · AI agents · MCP (Model Context Protocol) · AI code review

Practices

Advanced
Release engineering · Incident response · TDD · Mentoring

WORK EXPERIENCE

Senior Software Engineer II at StyleSeat

Feb 2014 - Jun 2026

Twelve years wearing many hats on a Python/Django marketplace: full-stack product, native mobile, search, payments, reliability, and most recently DevOps and platform engineering. Grew into a senior engineer the company relied on for its hardest, most cross-cutting problems.

- **AI code review:** Built "StyleSeatBot," an org-wide automated code-review bot powered by Claude. It became a required reviewer on every pull request, signing off alongside a human approver, with adoption up to the CTO.
- **AI enablement:** Helped drive the company's move to AI-assisted engineering as a member of the AI-adoption tiger team, running internal talks and workshops on Claude Code and LLM fundamentals.
- **Payments:** Longtime key payments-platform engineer and the team's payments domain expert; major contributor to the Stripe billing and subscriptions build-out (promotion codes, plan and product backfills,

coupon-stacking prevention, Stripe Portal, Pro Earnings tab).

- **Fraud detection:** Architected and built an automated fraud-detection system on Stripe Radar.
- **Platform migrations:** Led the pdm-to-uv migration; contributed to org-wide Python 2-to-3 (~900K LOC), Segment-to-Braze, and MySQL 5.7-to-8 migrations.
- **Reliability:** Owned a custom fault-tolerant Redis client library for graceful degradation when Redis was down, and built an S3-backed idempotent Celery task-recovery toolkit after a broker outage failed 351 tasks.
- **Release engineering:** Owned release tooling: a custom CircleCI orb, deploy-gate and hotfix/cherry-pick workflows, and build-failure alerting; contributed to CI/CD and Terraform/ECS.
- **Scale & mentorship:** Among the highest PR-review volume on the team; authored ~165K LOC across 180 PRs in 26 repos, and mentored engineers.

Software Engineer at Switchcam

Aug 2012 - Feb 2014

📍 San Francisco, CA / Chicago, IL

- Full-stack product engineering at a social-video startup, building Python/Django backends and JavaScript frontends across data models, APIs, and UI.

Lead Django Developer at Fulton Works

Feb 2011 - Aug 2012

📍 Chicago, IL

- Led Django development of client web applications at a development studio, owning architecture and full-stack delivery across multiple engagements.

Software Engineer at Hagerty Consulting

May 2009 - Feb 2011

📍 Evanston, IL

- Built internal web applications and data tooling in Python/Django to support consulting engagements.

Software Engineer / IT Engineer at MSDSONline

May 2006 - May 2009

📍 Chicago, IL

- Developed web application features for a SaaS chemical-safety and compliance platform, moving into software engineering from an IT and internal-tooling role.

PROJECTS

lockfreequeues

Skills: Nim · Lock-free · Concurrency

🔗 <https://github.com/elijahr/lockfreequeues>

Flagship Nim project: a library of eight lock-free queues (SPSC/SPMC/MPSC/MPMC, bounded and unbounded) with a uniform API and verified ordering/progress guarantees. 50+ stars, in the Nimble registry.

nim-debra

Skills: Nim · Memory reclamation · Systems

🔗 <https://github.com/elijahr/nim-debra>

DEBRA+ epoch-based memory reclamation for lock-free Nim data structures; the never-blocking engine underneath lockfreequeues. Nimble package, 15 releases.

python-autopxd2

Skills: Compiler tooling · libclang · Cython · C/C++

🔗 <https://github.com/elijahr/python-autopxd2>

Primary maintainer (inherited). Rewrote it from a limited pycparser tool into a full libclang-based one that generates Cython .pxd for any C or C++ clang can parse, and factored its parsing core into a reusable toolkit (headerkit). 52+ stars, on PyPI.

run-on-arch-action (major contributor)

Skills: CI/CD · QEMU · Cross-architecture · GitHub Actions

 <https://github.com/uraimo/run-on-arch-action>

Major contributor to a widely used GitHub Action (747 stars) that runs CI on non-x86 CPUs (ARM, aarch64, s390x, ppc64le, riscv64) via QEMU; refactored and rewrote it to be more maintainable, testable, and extensible, reviving active community contribution.

pynng (contributor)

Skills: Low-level protocols · Messaging · nng · Python

 <https://github.com/codypiersall/pynng>

Active contributor to the Python bindings for nanomsg-NG (nng), a high-performance low-level messaging and protocol library. 328 stars.

ringbuf

Skills: Realtime audio · DSP · Cython · Lock-free

 <https://github.com/elijahr/ringbuf>

Lock-free single-producer/single-consumer ring buffer for Python and Cython, built for realtime DSP where the audio callback must not allocate, lock, or touch the GIL. On PyPI.

axiomantic / spellbook

Skills: AI tooling · MCP · LLM · Developer experience

 <https://github.com/axiomantic/spellbook>

Multi-platform harness for AI coding assistants (skills, hooks, and a shared MCP server) across Claude Code, OpenCode, Codex, Gemini CLI, and ForgeCode, plus a family of MCP servers. Anchor of the axiomantic AI-tooling org. 100 releases.

EDUCATION

Bachelor of Science Computer Science, Minor in Philosophy at	Aug 2004 - May 2008
Northeastern Illinois University	

LANGUAGES

English	Spanish
Native speaker	Limited working proficiency